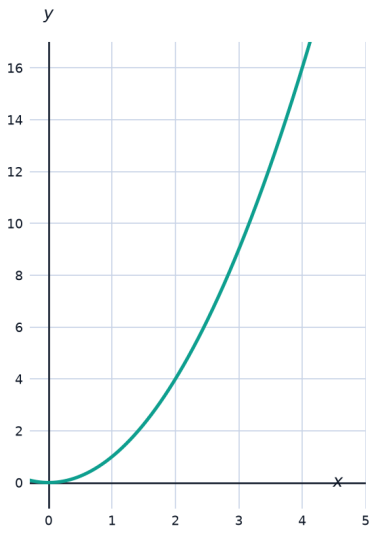
Riemann Sums and Definite Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

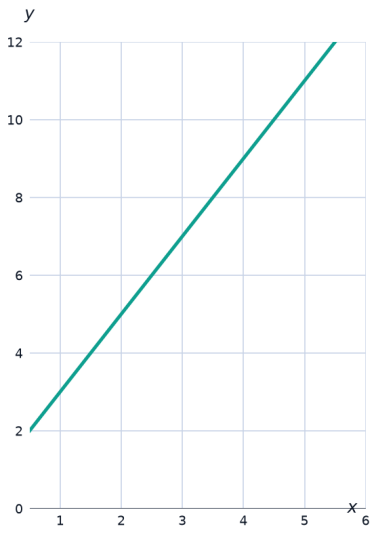
1 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_0^4 x^2\,dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.



2 [Foundational] • Target: Approximating area with Riemann sums

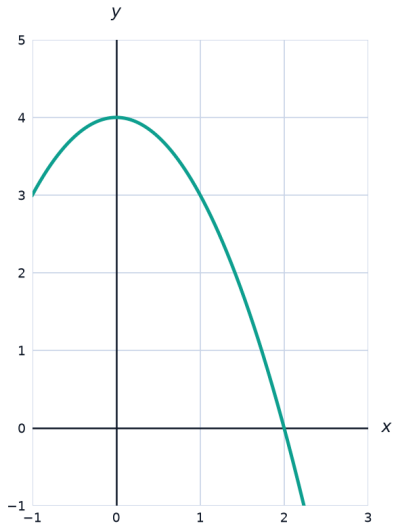
Estimate $\int_1^5 (2x + 1)\,dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.



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3 [Foundational] • Target: The definite integral and its properties

The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2)\,dx$ exactly.



4 [Foundational] • Target: The definite integral and its properties

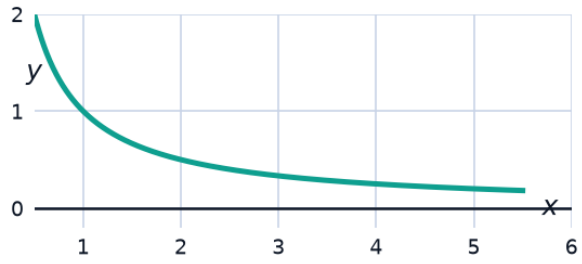
Evaluate $\int_{-2}^{2} (x^3 - 4x)\,dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

5 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_0^4 x^2\,dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.

6 [Foundational] • Target: Midpoint and trapezoidal approximations

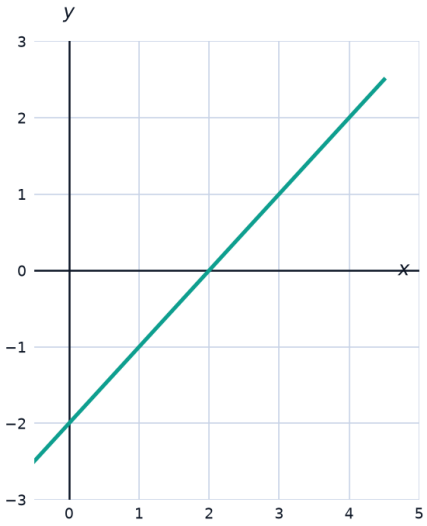
Estimate $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.



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7 [Foundational] • Target: Signed area

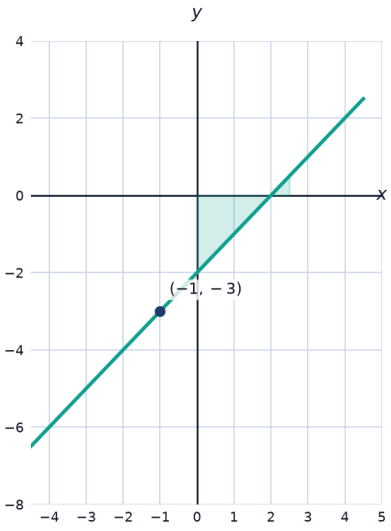
The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x - 2)\,dx$ and explain why the result does not equal the total area between the curve and the axis.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Signed area

Find the total (unsigned) area between $f(x) = x - 2$ and the x-axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.



9 [Intermediate] • Target: Comparing Riemann sum estimates

For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

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10 [Intermediate] • Target: Comparing Riemann sum estimates

Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

11 [Intermediate] • Target: Riemann sums from a table of values

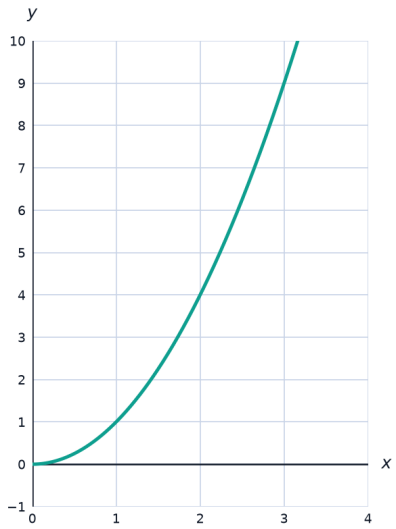
A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

12 [Intermediate] • Target: The definite integral as a limit of Riemann sums

Write the formal limit definition of the definite integral $\int_a^b f(x)\,dx$ in terms of a Riemann sum, and explain the role of $n \to \infty$.

13 [Intermediate] • Target: Riemann sums with sigma notation

Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2\,dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.



14 [Intermediate] • Target: Riemann sums with sigma notation

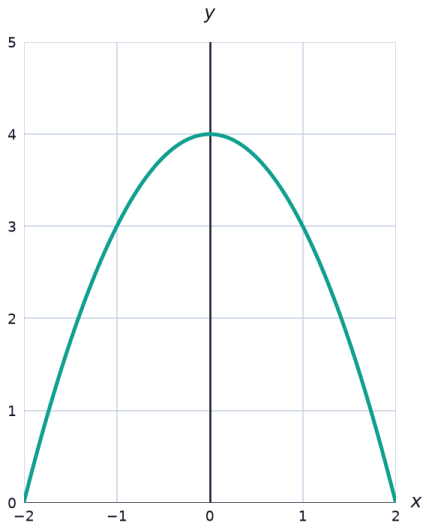
Estimate $\int_0^6 f(x)\,dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.



16 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

17 [Challenging] • Target: Riemann sums with sigma notation • Cross-Lesson: 15.01 Double Integrals

Level 1 · FoundationsExpress the following limit as a definite integral with lower limit $a = 0$, and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2}{n}\sqrt{4 - \left(\frac{2i}{n}\right)^2}$

18 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

19 [Challenging] • Target: Approximating area with Riemann sums • Cross-Lesson: 2.01 Evaluating Limits

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

20 [Challenging] • Target: The definite integral and its properties • Cross-Lesson: 2.02 Continuity

Level 1 · FoundationsEvaluate the improper integral $\int_1^{\infty} \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Approximating area with Riemann sums

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:

a. $n \to \infty$ (partition width $\to 0$)

b. $b \to \infty$

c. $f \to 0$

d. $x \to a$

22 [Foundational] • Target: The definite integral and its properties

The midpoint rule approximation for $\int_0^1 \int_0^1 f(x, y)\, dA$ with one rectangle:

a. $f(1, 1)$

b. $f(0, 0)$

c. $f(1/2, 1/2)$

d. $f(0, 1)$

23 [Foundational] • Target: Midpoint and trapezoidal approximations

Integral test: $\sum_{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)dx$ converges, for f positive decreasing. So $\sum 1/n^2$ converges because $\int_1^{\infty} x^{-2}dx =$

a. 1 (finite)

b. ∞

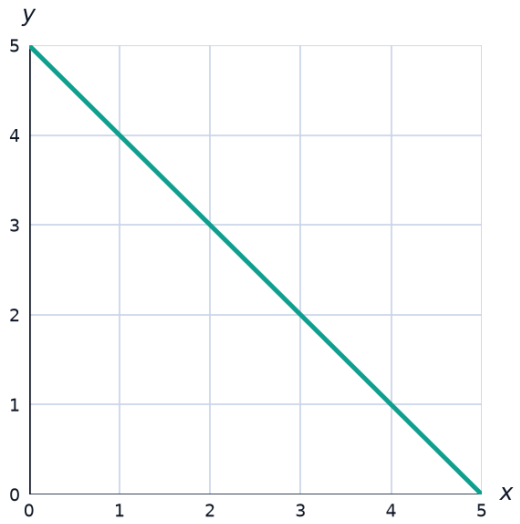
c. 0

d. 1/2

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24 [Foundational] • Target: Signed area

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?



a. 4.5

b. 6

c. 7.5

d. 9

25 [Foundational] • Target: Comparing Riemann sum estimates

Evaluate the improper integral $\int_0^\infty \frac{1}{x^2+9}\,dx$.

a. $\frac{\pi}{3}$

b. ∞

c. $\frac{\pi}{6} + 1$

d. $\frac{\pi}{6}$

26 [Foundational] • Target: Riemann sums from a table of values

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s?

a. 8 m²/s

b. 16 m²/s

c. 32 m²/s

d. 4 m²/s

27 [Foundational] • Target: The definite integral as a limit of Riemann sums

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2+y^2} =$

a. always 1

b. an integer

c. always 0

d. always π

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28 [Foundational] • Target: Riemann sums with sigma notation

Triple integral $\iiint_V 1\,dV$ gives:

a. volume of V

b. 0

c. surface area

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing approximation methods on a concave function

Riemann sum step width $\Delta x =$

a. $1/n$

b. $(b-a)/n$

c. $n(b-a)$

d. $(f(b)-f(a))/n$

30 [Intermediate] • Target: Approximating area with Riemann sums

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

31 [Intermediate] • Target: The definite integral and its properties

A rectangle has perimeter 40. Find the width w that maximises its area.

a. 10

b. 20

c. 5

d. $\frac{40}{3}$

32 [Intermediate] • Target: Midpoint and trapezoidal approximations

$\int_0^1 \int_0^x y\,dy\,dx$: inner integral gives $x^2/2$, so outer integral $=$

a. 1

b. 1/4

c. 1/2

d. 1/6

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33 [Intermediate] • Target: Signed area

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region S in the uv-plane with area 2 maps to a region R. Find the area of R.

a. 2

b. 8

c. 12

d. 16

34 [Intermediate] • Target: Comparing Riemann sum estimates

Green's theorem area: Area $= \iint_D dA = \frac{1}{2}\oint_C (x\,dy - y\,dx)$. For unit circle:

a. π

b. 2π

c. 1

d. $\pi/2$

35 [Intermediate] • Target: Riemann sums from a table of values

A deltoid (3-cusp hypocycloid): $x = \frac{2}{3}a\cos t + \frac{a}{3}\cos 2t$, $y = \frac{2}{3}a\sin t - \frac{a}{3}\sin 2t$. Its area:

a. $\pi a^2/4$

b. $\pi a^2/3$

c. $2\pi a^2/3$

d. πa^2

36 [Intermediate] • Target: The definite integral as a limit of Riemann sums

Green's theorem area formula: area $= \frac{1}{2}\oint_C (x\,dy - y\,dx)$. For ellipse $x = a\cos t$, $y = b\sin t$:

a. $\pi(a + b)$

b. $2\pi ab$

c. πab

d. ab

37 [Intermediate] • Target: Riemann sums with sigma notation

Generating function of Fibonacci: $F(x) = \sum_{n=0}^{\infty} F_n x^n = \frac{x}{1 - x - x^2}$. The coefficient of x^6:

a. 13

b. 8

c. 5

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 16.01 Vector Fields and Line Integrals

For $|x| < 1$, which function equals $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$?

a. $\frac{1}{1-x}$

b. $\frac{1}{(x+1)^2}$

c. $\ln(x+1)$

d. $-\ln(1-x)$

39 [Challenging] • Target: Approximating area with Riemann sums • Cross-Lesson: 2.01 Evaluating Limits

A solid cylinder $x^2 + y^2 \le 16,\ 0 \le z \le 4$ has density $\delta = \sqrt{x^2 + y^2}$. Find its mass.

a. 64π

b. $\frac{1024\pi}{3}$

c. $\frac{256\pi}{3}$

d. $\frac{512\pi}{3}$

40 [Challenging] • Target: The definite integral and its properties • Cross-Lesson: 2.02 Continuity

Laurent series: a function analytic in an annulus $r < |z| < R$ has a series $\sum_{n=-\infty}^{\infty} a_n z^n$. For $1/(z(1-z))$ near $z = 0$:

a. $1/z^2 + 1/z + \cdots$

b. $1/z - 1 + z - \cdots$

c. $z + z^2 + \cdots$

d. $1/z + 1 + z + z^2 + \cdots$

41 [Challenging] • Target: Midpoint and trapezoidal approximations • Cross-Lesson: 3.01 The Derivative from First Principles

A lamina occupies $[0,3] \times [0,3]$ with density $\rho = xy$. Find its mass.

a. $\frac{81}{2}$

b. $\frac{81}{4}$

c. $\frac{81}{8}$

d. 9

42 [Challenging] • Target: Signed area • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Area of disk $x^2 + y^2 \le R^2$ by polar coordinates:

a. πR

b. $2\pi R$

c. πR^2

d. $4\pi R^2$

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43 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

$\int_0^2 \int_0^{2-x} dy\,dx$: region is a triangle. Area $=$

a. 1

b. 4

c. 2

d. 3

44 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Apply Fubini: $\int_0^\infty \int_0^\infty e^{-xy}\sin x\,dy\,dx$. Inner integral: $\int_0^\infty e^{-xy}dy =$

a. x

b. e^{-x}

c. $1/x$

d. 1

45 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 6.01 Related Rates

The Euler-Mascheroni constant $\gamma = \lim_{n\to\infty}(H_n - \ln n) \approx 0.5772$ where $H_n = \sum_{k=1}^{n} 1/k$. Its rationality or irrationality:

a. known rational

b. known irrational

c. zero

d. unknown (open problem)

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